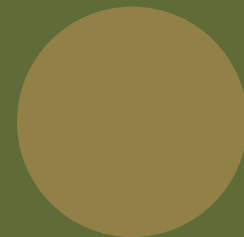


Global CO₂ Emissions

Trends, Regional Shifts, and
Per Capita Inequalities

A Data-Driven Overview: 1750 – 2024

Data: Global Carbon Budget 2025 | NASA Vital Signs



429

ppm CO₂ (Feb 2026)

An orange pill-shaped graphic with rounded ends, containing the text "1750 – 2024".

1750 – 2024

Atmospheric CO₂ Reaches 429 ppm

50% Above Pre-Industrial Levels

429

parts per million

February 2026 — Mauna Loa Observatory

+50% from pre-industrial

Context & Acceleration

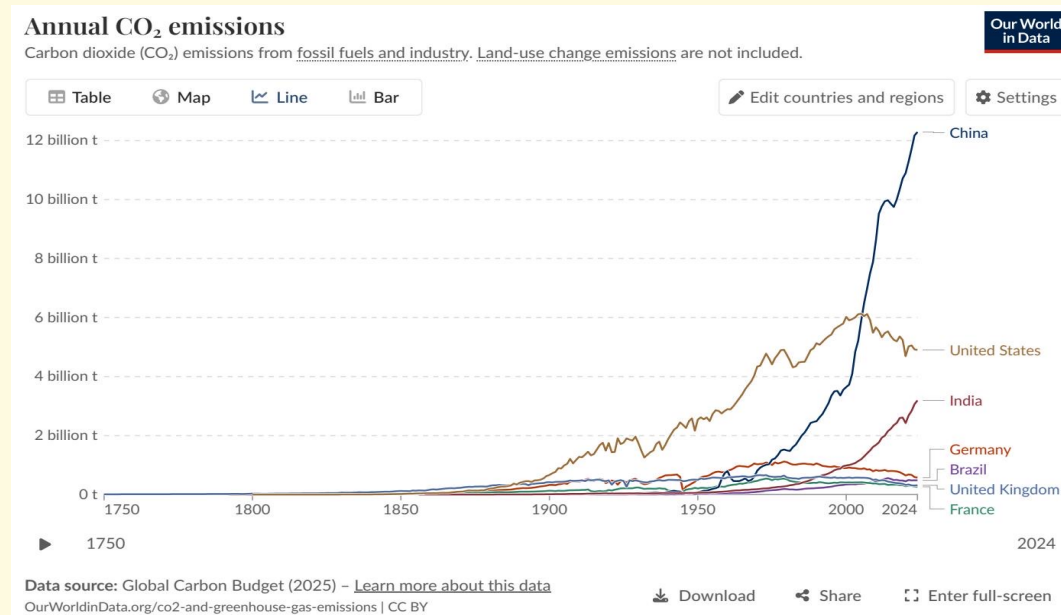
~1750 Pre-industrial baseline: ~280 ppm

1958 Keeling Curve begins: ~315 ppm

2026 Current level: 429 ppm
The current increase is at least 1.5× the natural rise at the end of the last ice age — a change that took thousands of years then but has occurred in roughly two centuries now.

Global Fossil Fuel CO₂ Emissions Grew Six-Fold

6 Billion to 35+ Billion Tonnes (1950–2024)



Key Milestones

1950

~6 Bt CO₂

1990

>20 Bt CO₂

2024

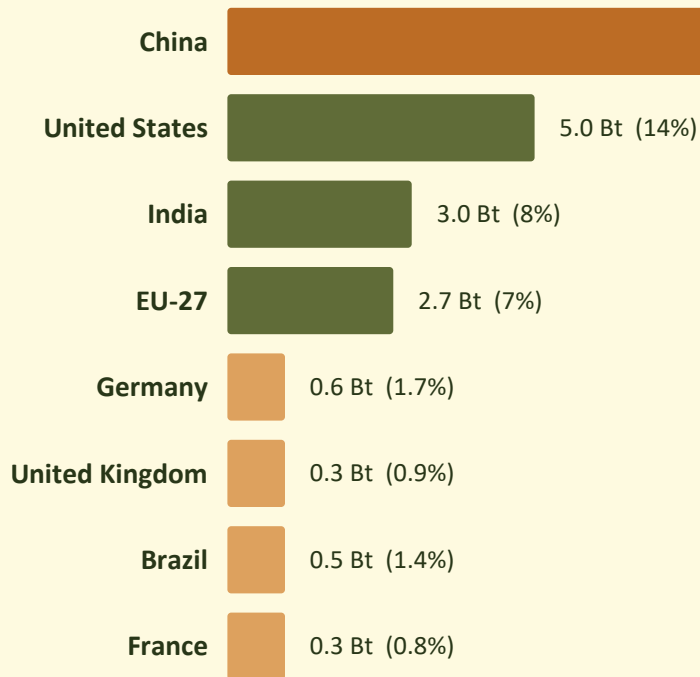
>35 Bt CO₂

Growth remained slow until the mid-20th century, then accelerated sharply. While growth has slowed in recent years, emissions have not yet peaked.

Source: Our World in Data — Global Carbon Budget 2025

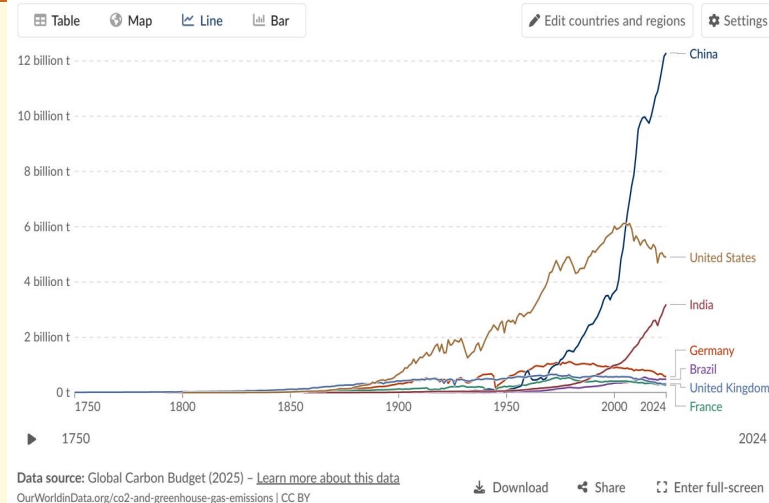
China Now Emits Over 12 Billion Tonnes

More Than Double the United States



Annual CO₂ emissions

Carbon dioxide (CO₂) emissions from fossil fuels and industry. Land-use change emissions are not included.



China's steep rise post-2000 contrasts with the US decline from its ~6 Bt peak around 2005.

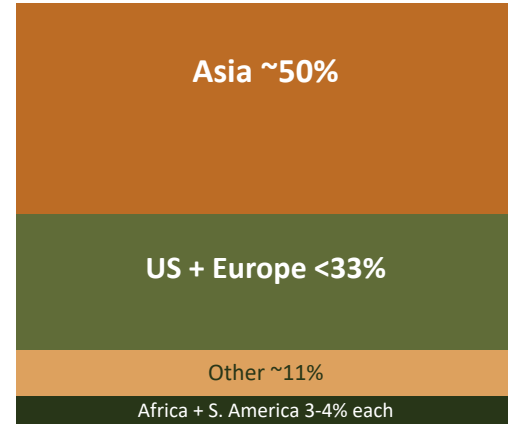
Asia Now Accounts for ~50% of Global Emissions

US + Europe Below One-Third

1900



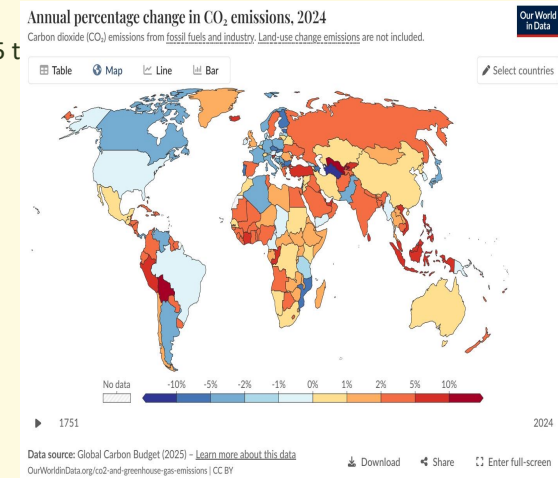
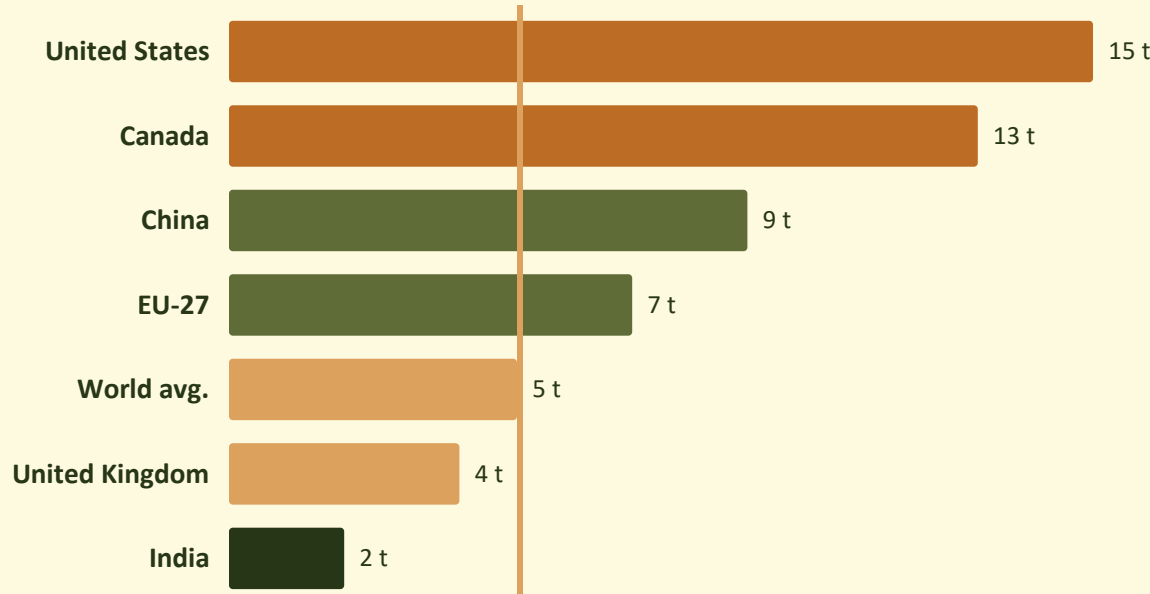
2024



In 1900, Europe and the US accounted for >90% of global emissions. By 1950, still >85%. Today, Asia — home to nearly 60% of the world's population — produces roughly half.

Per Capita Gap: 150× Disparity

15 t in the US vs. 0.1 t in Chad

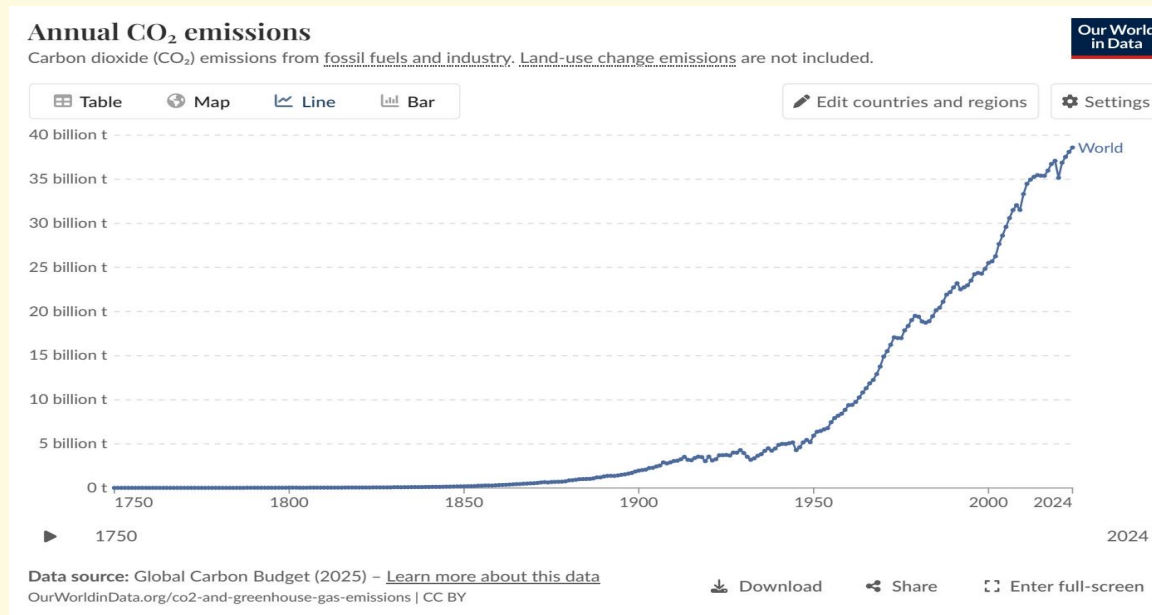


Per capita CO₂ emissions over time

The average American emits in under two days what the average person in Chad or Niger emits in an entire year. Countries with similar living standards can differ significantly — France and the UK have much lower per capita emissions than Germany or the Netherlands.

2024 Snapshot: Mixed Global Trajectories

US and Europe Cut Emissions While Russia and SE Asia Increase



Annual percentage change in CO₂ emissions, 2024

Regional Patterns

Declining

US, much of Europe
–1% to –5%

Increasing

Russia, SE Asia, parts of Africa
+2% to +10%

Sharp increase

Some South American nations
>10% increase

Mixed

Some S. America declined
while others rose sharply

Energy Mix Explains Per Capita Differences

France and the UK emit far less per capita than Germany or the Netherlands

Country	Nuclear	Renewables	Fossil Fuels	Per Capita CO ₂
France	High	Moderate	Low	~4 t
United Kingdom	Moderate	Growing	Declining	~4 t
Germany	Low (phase-out)	Growing	Higher	~8 t
Netherlands	Low	Moderate	Higher	~8 t

France & UK

Higher nuclear and renewable shares drive per capita emissions to ~4 t — roughly half that of Germany and the Netherlands.

Germany & Netherlands

Higher reliance on fossil fuels (especially coal and gas) pushes per capita emissions to ~8 t — despite comparable prosperity levels.

Key Insight: Policy and energy mix choices — not just prosperity — determine per capita outcomes.

Key Takeaways

Emissions Are Still Rising, But the Map of Responsibility Is Shifting

- 1 Global CO₂ emissions exceed 35 billion tonnes per year and have not yet peaked, despite slowing growth rates.
- 2 China alone accounts for over 25% of global emissions (~12.5 Bt), more than double the United States (~5 Bt); India (~3 Bt) is the third-largest and rising fast.
- 3 The geographic center of emissions has shifted dramatically: from >90% in Europe and the US in 1900 to Asia producing ~50% today.
- 4 Per capita inequalities remain extreme — a 150× gap separates the highest emitters (US, Australia ~15 t/person) from the lowest (Chad, Niger ~0.1 t/person).
- 5 Policy and energy mix choices — not just prosperity — determine per capita outcomes. France and the UK emit half as much per capita as Germany or the Netherlands.